

Student Enrollment & Course Management System (SECMS)

User Story:

As an enrollment officer, I want to manage student enrollment and course details so that I can process registrations quickly and accurately. I want to register new students, update their information, manage course details, and process enrollment requests. I also want to track enrollment status, record fee payments, and receive approval updates. This system helps reduce manual work, avoid duplicate entries, improve data accuracy, and make the enrollment process faster and easier.

1.Introduction

1.1 Project Overview

This project is designed to bring student, course, and enrollment information together in a single Salesforce application, providing easy access to data, organized workflows, and accurate reporting. It enables Enrollment Officers to efficiently manage student admissions, course registrations, and communication, while giving instructors and administrators clear visibility into course assignments, student enrollments, and academic progress.

The system also simplifies enrollment management through **automated instructor, assignment, task creation, data validation, and multi-level approval processes**. These features help reduce manual work, minimize errors, and ensure that enrollment activities are completed smoothly and efficiently.

By using **Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards**, the organization can manage the complete enrollment journey in a structured and systematic manner from student admission to course completion. The system also provides valuable insights into key areas such as **enrollment volume, revenue, student**

status, course participation, and pending approvals, helping administrators monitor performance and make informed decisions.

1.2 Purpose

Purpose of the System

The main purpose of SECMS is to make the student enrollment process simple, fast, organized using Salesforce.

The system helps to Store student and course details in one place.

- Store student and course details in one place.
- Manage student enrollments easily.
- Automate approvals and validations.
- Reduce manual work and errors.
- Track enrollment activities.
- Generate simple reports and dashboards.
- Improve communication between staff and instructors.

2. Ideation Phase

2.1 Problem Statement (Customer Problem Statement Template)

PS-1: Enrollment Officer

The Enrollment Officer needs to manage student enrollments efficiently, but the data is scattered and the process is mostly manual, which makes the work difficult and prone to errors.

PS-2: Administrator

The Administrator needs to track enrollments and revenue, but the data is not centralized and reports are delayed, making it difficult to make quick and accurate decisions.

PS-3: Instructor

The Instructor needs to view assigned courses and student details, but the information is not easily available, making course management more difficult.

PS-4: Enrollment Officer

The Enrollment Officer needs to process enrollments quickly, but manual approval processes cause delays and make the enrollment process less efficient.

2. Ideation Phase



Fig 2.1 — Customer problem statement canvas

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- ❖ Think & Feel: Worried about duplicate bookings and payment issues.
- ❖ Hear: Feedback and complaints from students and management.
- ❖ See: Different spreadsheets, emails, and student records.
- ❖ Say & Do: Checks information and follows up with students manually.
- ❖ Pain: Duplicate data, delays in approvals, and manual work.
- ❖ Gain: Automated processes, clear dashboards, and faster enrollment management.

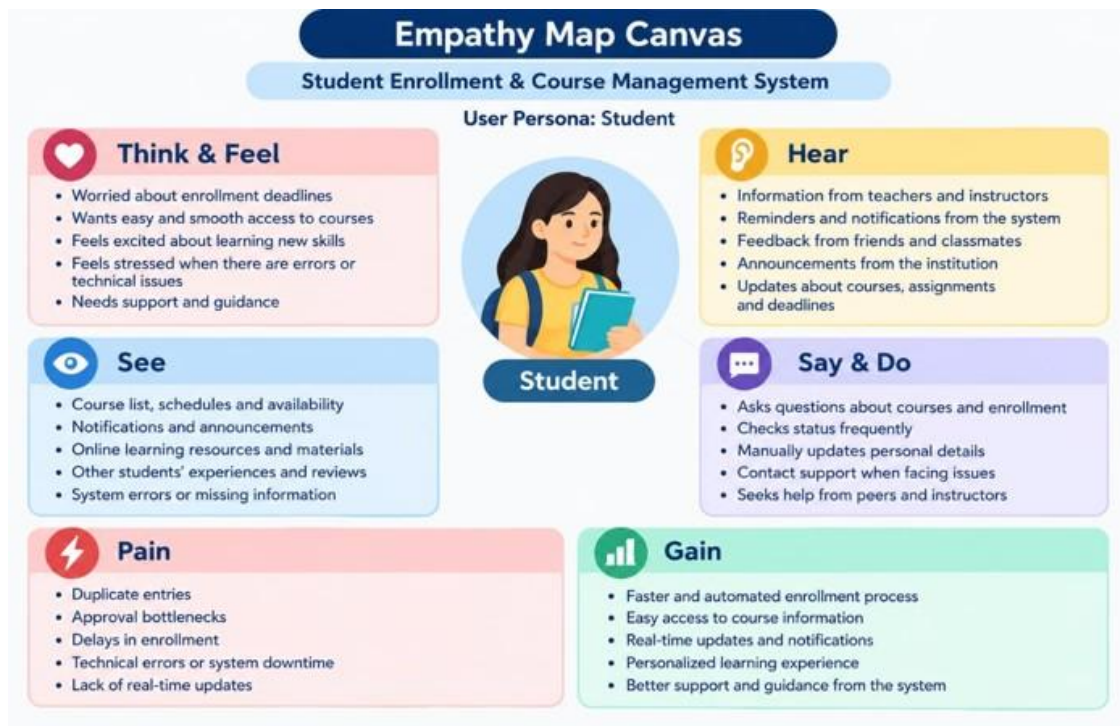


Fig 2.2 — Empathy map canvas for the Enrollment Officer persona

Brainstorming & Idea Prioritization

Step-1 Brainstormed ideas

- Step Spreadsheet-based automation
- Standalone student enrollment portal
- ERP system integration
- Salesforce-based enrollment management system

Step 2 – Grouping of Ideas

- Manual-based solutions

- Semi-automated solutions
- Fully automated cloud-based solutions

Step 3 – Idea Prioritization

- ❖ The ideas were prioritized based on automation, scalability, ease of use, cost, and integration capabilities.
- ❖ Salesforce-based Enrollment Management System – High Priority
- ❖ ERP Integration – Medium Priority
- ❖ Standalone Enrollment Portal – Medium Priority
- ❖ Spreadsheet Automation – Low Priority

2.3 Brainstorming & Idea Prioritization

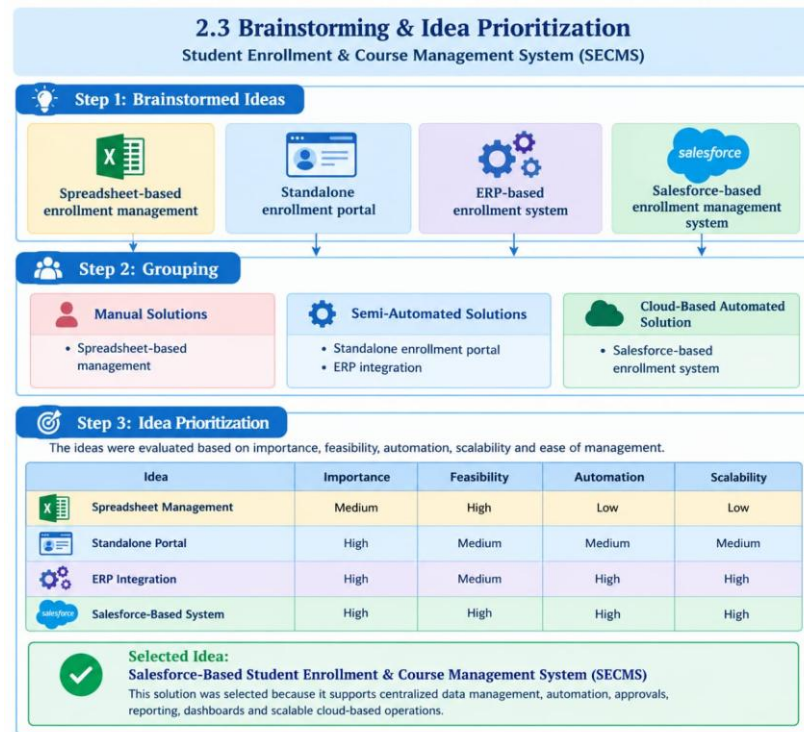


Fig 2.3 — Brainstorming, grouping, and prioritization of solution ideas

For your SECMS project, you can write it like this:

1. Grouping: The solution ideas were grouped into manual, semi-automated, and fully automated solutions.
2. Prioritization: The ideas were prioritized based on cost, ease of implementation, scalability, automation, and user benefits.

3. Selected Solution: The Salesforce-based enrollment system was selected as the preferred solution because it provides centralized data management and automation.

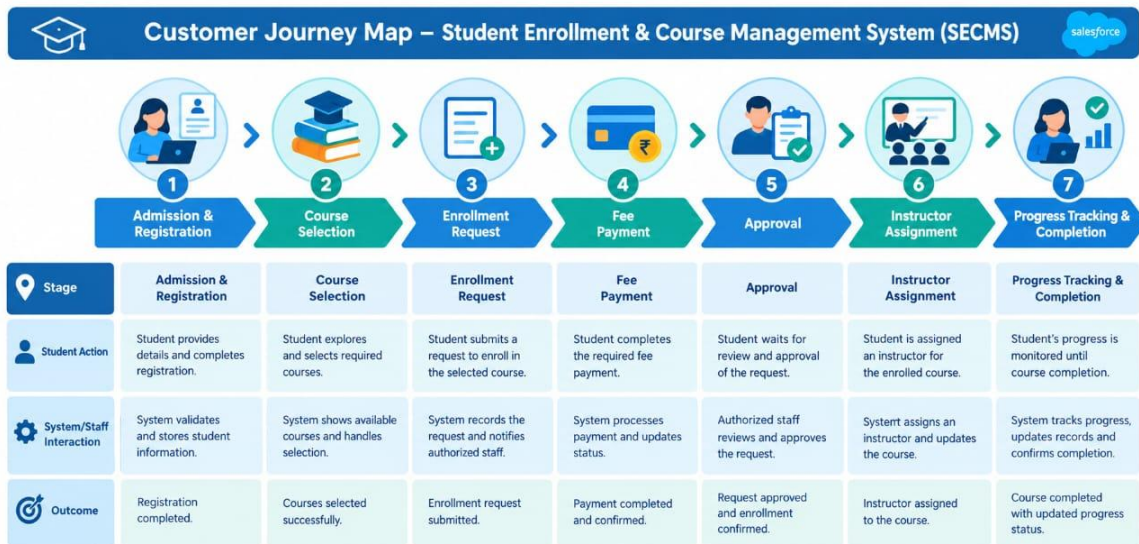
2.4 REQUIREMENT ANALYSIS PHASE

2.5 Customer Journey Map

Stages cover:

For your Student Enrollment & Course Management System (SECMS), you can change the sentences like this:

- ❖ Admission & Registration – Students provide their details and complete the registration process.
- ❖ Course Selection – Students explore and select the required courses.
- ❖ Enrollment Request – Students submit a request to enroll in the selected course.
- ❖ Fee Payment – Students complete the required fee payment.
- ❖ Approval – The enrollment request is reviewed and approved by the Instructor
Assignment – An instructor is assigned to the enrolled course.
- ❖ Instructor Assignment – An instructor is assigned to the enrolled course.
- ❖ Progress Tracking & Completion – Student progress is monitored until successful course completion.



2.6 Solution Requirements

Functional Requirement

FR-1: Student Management

The system shall allow administrators to create, view, update, and manage student records.

FR-2: Course Management

The system shall allow authorized users to create, update, view, and manage course details.

FR-3: Enrollment Processing

The system shall allow users to submit, approve, or reject student enrollment requests.

FR-4: Instructor Management

The system shall allow administrators to create, assign, and manage instructor details and course assignments.

FR-5: Payment Management

The system shall allow users to record, track, and manage student fee payment details.

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple and user-friendly interface that allows users to perform tasks easily with minimal training.

- Easy student registration
- Simple enrollment process
- Clear and role-based dashboards

NFR-2: Security

The system shall protect student and institutional data through secure authentication and role-based access control.

- Only authorized users can access the system
- Only administrators can modify course details
- Instructors can view only assigned student records
- Students cannot access other students' records

NFR-3: Reliability

The system shall work consistently and accurately without errors during enrollment, approval, and report generation. It shall prevent duplicate or incorrect course enrollments.

- Prevent duplicate course registration.
- Update enrollment status accurately after approval.
- Generate reports consistently and accurately.

NFR-4: Performance

The system shall provide fast response times for all operations and ensure smooth system performance, especially during peak admission periods.

- Quickly load dashboards and reports.

- Process enrollment requests without unnecessary delays.

NFR-5: Scalability

The system shall support an increasing number of students, courses, departments, and users without affecting system performance. It should also allow new modules and features to be added easily.

- Add new departments without redesigning the system.
- Handle increased student admissions every year.
- Support future mobile or portal integration.

NFR-6: Availability

The system shall be available to users during working hours and minimize down time. It should provide reliable access for academic and administrative activities.

- Provide 24/7 access to the system.
- Minimize system downtime.
- Ensure reliable access for users.

2.7 Data Flow Diagram

- Users enter student, course, and enrollment details into the system.
- The system checks the entered information and saves it in Salesforce.
- Enrollment requests are sent for approval based on the required process.
- The system assigns instructors to approved courses or enrollments.
- The processed information is used to create reports and dashboards.
- Administrators can use these reports to monitor enrollments and overall perform.

Overall, the system collects enrollment information, checks the data, manages approvals, assigns instructors, and provides useful reports for monitoring.

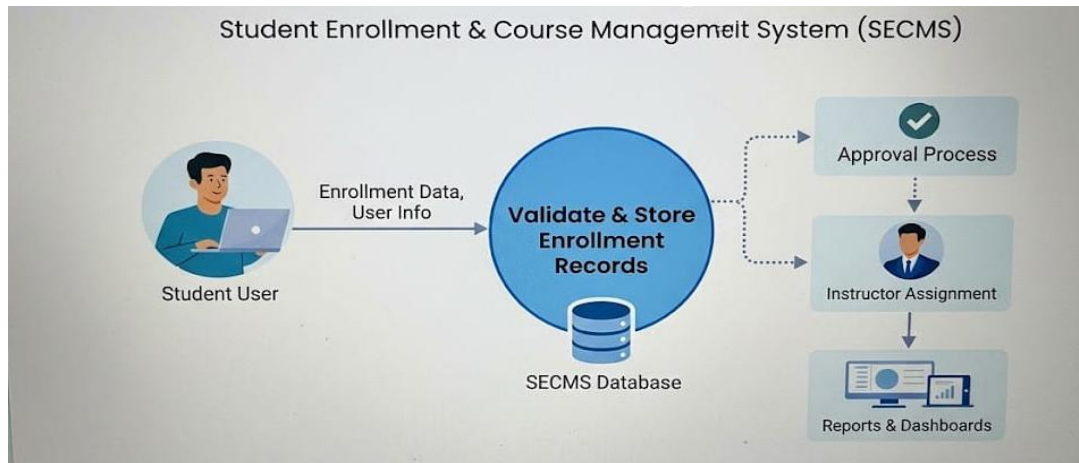


Fig 2.7— Data flow diagram for SECMS

2.8 Technology Stack

The application's architecture utilizes Salesforce Lightning for the UI layer, Salesforce Flows for business logic, Salesforce Objects for the database, Validation Rules and Approval Processes for automation, and Reports & Dashboards for reporting.

- ❖ **UI:** Built using Salesforce Lightning.
- ❖ **Logic:** Powered by Salesforce Flows.
- ❖ **Database:** Structured using Salesforce Objects.
- ❖ **Automation:** Handled via Validation Rules and Approval Processes.
- ❖ **Reporting:** Delivered through Reports & Dashboards.

2.9 Project Design Phase

3 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

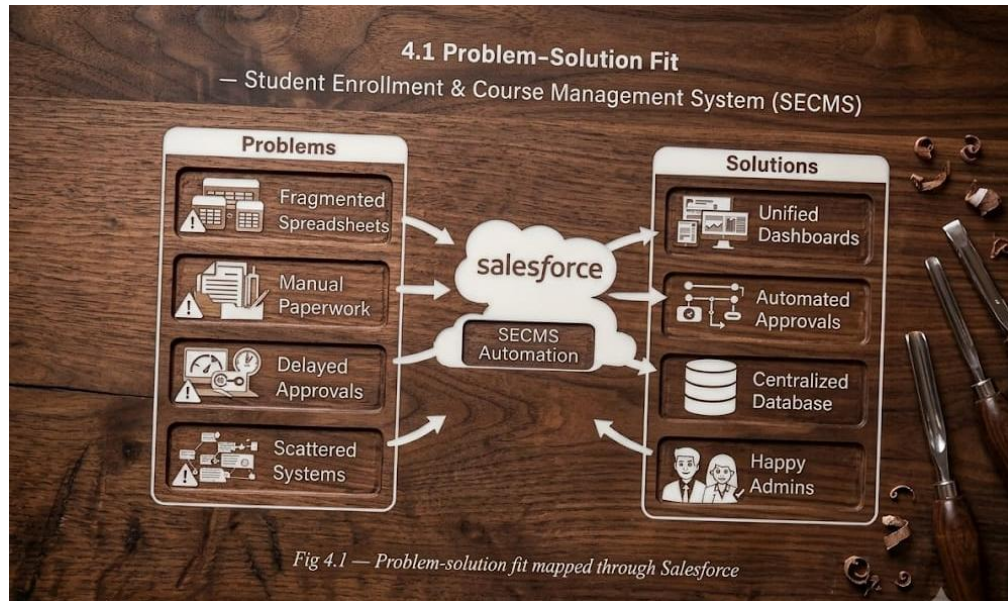


Fig 3 — Problem–solution fit mapped through Salesforce

3.1 Proposed Solution

Problem: The problem is manual enrollment handling.

Solution: The solution is Salesforce-based SECMS.

Novelty: The novelty lies in end-to-end automation.

Social Impact: The social impact is an improved student experience.

Revenue Model: The revenue model is based on institutional deployment.

Scalability: Scalability is achieved through multi-department support.

3.2 Solution Architecture

The system is built on Salesforce and combines data management, automation

Reporting and security features to support the enrollment process.

- ❖ **Objects:** Manage Student, Course, Enrollment, and Instructor information.
- ❖ **Automation:** Uses Flows and Approval Processes to automate tasks and approvals.
- ❖ **Reporting:** Uses Reports and Dashboards to monitor enrollment activities and performance.
- ❖ **Security:** Uses Profiles, Permission Sets, and Sharing Rules to control data access.

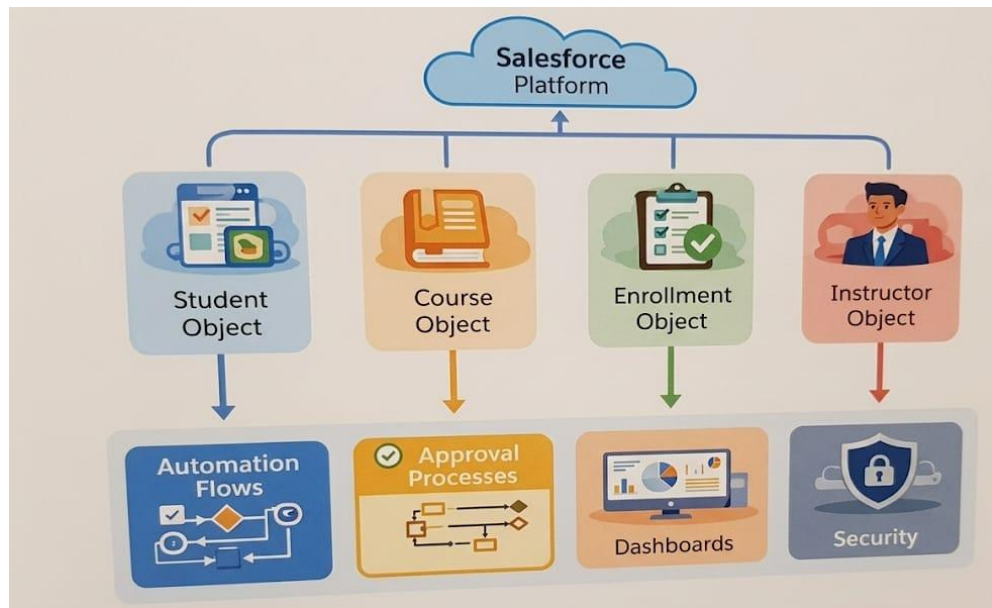


Fig 3.2 — Solution architecture: objects, automation, and security layers

3.3 Project Planning & Scheduling

The project follows an **Agile approach**, where the work is divided into **Epics, User Stories, and Story Points**. Development is completed in **Sprints**, with the **Product Backlog** used to manage tasks, the **Sprint Schedule** used to plan activities, and **Velocity** used to estimate the work completed in each sprint.

The **Student Enrollment & Course Management System** follows an **Agile methodology**. The project is divided into small tasks and completed through **sprints**.

- **Planning:** Identify project requirements and create the Product Backlog.
- **Estimation:** Use Story Points to estimate task effort.
- **Scheduling:** Plan tasks and activities for each sprint.
- **Execution:** Complete development sprint by sprint.
- **Tracking:** Use Sprint Reviews and Velocity to monitor progress.

3.4 Project Development Phase

Project Flow

1. Milestone 1: Salesforce Developer Account Creation
2. Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
3. Milestone 3: Tabs Creation
4. Milestone 4: Lightning App Creation
5. Milestone 5: Fields Creation
6. Milestone 6: Create Relationships
7. Milestone 7: Page Layout Customization
8. Milestone 8: Validation Rules
9. Milestone 9: Flows (Automation)
10. Milestone 10: Approval Process
11. Milestone 11: Reports
12. Milestone 12: Dashboards
13. Milestone 13: Security Setup
14. Milestone 14: Conclusion

What You'll Learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards

8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing.

Activity 1: Creating Developer Account

Creating a developer org in Salesforce. Go to <https://developer.salesforce.com/signup> and enter the following details on the sign-up form:

- First name & Last name
- Email
- Role: Developer
- Company: College Name
- Country: India
- Postal Code: Pin code
- Username: a combination of name and company (need not be an actual email id, in the format `username@organization.com`)

Click Sign Me Up after filling these details.

Fig 3.5— Salesforce Developer Edition sign-up page

3.6 FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features within the Student Enrollment & Course Management System (SECMS). These features include Record Creation: Successful creation of student, course, instructor, and enrollment records Validation Rules: Verification of mandatory fields, eligibility criteria, and enrollment. Evaluation of system behavior across different course categories. Process Automation: Flow execution for automated enrollment approvals and seat availability checks Apex Triggers: Verification of trigger functionality to ensure automatic confirmations and real-time record updates. Additionally, screenshots of generated reports and dashboards were documented to validate accurate data visualization, enrollment tracking, and performance monitoring. All screenshots captured during testing must be submitted along with this report. Functional & Performance Testing

3.7 Advantages & Disadvantages Data:

Advantages

- **Centralized Data:** All student and course information is stored in one place.
- **Automation:** Reduces manual work through automated workflows.
- **Real-Time Reports:** Provides quick access to updated reports and dashboards.

Disadvantages

- ❖ **Salesforce Dependency:** The system depends on the Salesforce platform.
- ❖ **Initial Setup:** Requires time and effort for configuration and customization.

3.8 Conclusion

The Student Enrollment & Course Management System (SECMS) provides a centralized and efficient way to manage student records, courses, enrollments, instructors, and approvals. It reduces manual work and errors through Salesforce automation, improving data accuracy and overall management efficiency. The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce's automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

3.9 Future Scope

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

4.0 Appendix

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub/ Demo Salesforce Developer Org

THANK YOU